

On the Compactness of Palais-Smale Sequences for p -Laplace Equations with Critical Nonlinearities

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ABSTRACT. In this talk, we will discuss a representation theorem of Mercuri-Willem (2010) for Palais-Smale sequences involving critical p -Laplace equations in smoothly bounded domains with negative parts vanishing at infinity.

1. Introduction

To begin, let us set up notation and context. Let $N \geq 3$, $1 < p < N$, and assume that Ω is a bounded domain in $\mathbb{R}^N$ having smooth boundary. We fix $a \in L^{N/p}(\Omega)$ and denote by p' and p^* the Hölder and Sobolev conjugate exponents of p , respectively. That is,

$$\frac{1}{p} + \frac{1}{p'} = 1 \quad \text{and} \quad p^* := \frac{Np}{N-p}.$$

We will be interested the p -Laplace operator:

$$\Delta_p u = \operatorname{div} \left(|\nabla u|^{p-2} \nabla u \right).$$

Before proceeding further, we require the notion of a homogeneous Sobolev space.

DEFINITION. Let $U \subseteq \mathbb{R}^N$ be open; $\mathcal{D}^{1,p}(U)$ is defined to be the real vector space of all functions $u \in L^{p^*}(U)$ such that $\nabla u \in L^p(U; \mathbb{R}^N)$. We give this space the seminorm

$$\|u\| := \|\nabla u\|_{L^p}.$$

Note that $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is actually a normed space.

If $U \subset \mathbb{R}^N$ is smoothly bounded, then $\mathcal{D}^{1,p}(U)$ and $W^{1,p}(U)$ consist of precisely the same functions. However, $\mathcal{D}^{1,p}(U)$ will possess a finer topology. When $U = \mathbb{R}^N$ however, we only have $W^{1,p}(\mathbb{R}^N) \subseteq \mathcal{D}^{1,p}(\mathbb{R}^N)$ by the Sobolev embedding theorem.

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Define on $W_0^{1,p}(\Omega)$

$$\phi(u) := \int_{\Omega} \left(\frac{|\nabla u|^p}{p} + a \frac{|u|^p}{p} - \mu \frac{|u|^{p^*}}{p^*} \right), \quad u \in W_0^{1,p}(\Omega),$$

which is the energy functional associated with the problem

$$\begin{cases} -\Delta_p u + a |u|^{p-2} u \equiv \mu |u|^{p^*-2} u & \text{in } \Omega, \\ u \equiv 0 & \text{on } \partial\Omega. \end{cases} \quad (1.1)$$

The weak formulation of the above is determined by the derivative of ϕ ,

$$\langle \phi'(u), h \rangle = \int_{\Omega} \left(|\nabla u|^{p-2} \nabla u \cdot \nabla h + a |u|^{p-2} u h - \mu |u|^{p^*-2} u h \right)$$

with $u, h \in W_0^{1,p}(\Omega)$. We will also require the energy functional

$$\phi_{\infty} : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad u \mapsto \int_{\mathbb{R}^N} \left(\frac{|\nabla u|^p}{p} - \mu \frac{|u|^{p^*}}{p^*} \right). \quad (1.2)$$

Hence, for $u, h \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ we get

$$\langle \phi'_{\infty}(u), h \rangle = \int_{\mathbb{R}^N} \left(|\nabla u|^{p-2} u h - \mu |u|^{p^*-2} u h \right).$$

Here, ϕ_{∞} is the energy functional associated with the limiting problem

$$-\Delta_p u \equiv \mu |u|^{p^*-2} u \quad \text{in } \mathbb{R}^N. \quad (1.3)$$

Our goal is to establish the following global compactness result due to Mercuri-Willem [5]:

THEOREM 1 (Mercuri-Willem, 2010). *Let (u_n) be a sequence in $W_0^{1,p}(\Omega)$ such that*

$$\phi(u_n) \rightarrow c \quad \text{and} \quad \phi'(u_n) \rightarrow 0 \quad \text{in } W^{-1,p'}(\Omega)$$

as $n \rightarrow \infty$. Assume further that

$$\lim_{n \rightarrow \infty} \|(u_n)_-\|_{L^{p^*}(\Omega)} = 0.$$

After passing to a subsequence, there exists a solution $v_0 \in W_0^{1,p}(\Omega)$ to the problem

$$\begin{cases} -\Delta_p u + a u^{p-1} = \mu u^{p^*-1} & \text{in } \Omega, \\ u \geq 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

and a finite (possibly empty) family $\{v_1, \dots, v_k\} \subset \mathcal{D}^{1,p}(\mathbb{R}^N)$ solving the limiting problem

$$\begin{cases} -\Delta_p u \equiv \mu |u|^{p^*-2} u & \text{in } \mathbb{R}^N, \\ u \geq 0 & \text{in } \mathbb{R}^N \end{cases} \quad (1.4)$$

together with associated sequences $(y_n^i)_{n \in \mathbb{N}} \subset \Omega$ and $(\lambda_n^i)_{n \in \mathbb{N}} \subset \mathbb{R}_+$ such that

$$\left\| u_n - v_0 - \sum_{i=1}^k (\lambda_n^i)^{\frac{p-N}{p}} v_i \left(\frac{\cdot - y_n^i}{\lambda_n^i} \right) \right\| \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (1.5)$$

$$\|u_n\|^p \rightarrow \sum_{i=0}^k \|v_i\|^p \quad \text{as } n \rightarrow \infty, \quad (1.6)$$

$$\phi(v_0) + \sum_{i=1}^k \phi_\infty(v_i) = \lim_{n \rightarrow \infty} \phi(u_n). \quad (1.7)$$

Moreover, there holds

$$\frac{\text{dist}(y_n^i, \partial\Omega)}{\lambda_n^i} \rightarrow \infty, \quad \text{as } n \rightarrow \infty$$

for each $i = 1, \dots, k$.

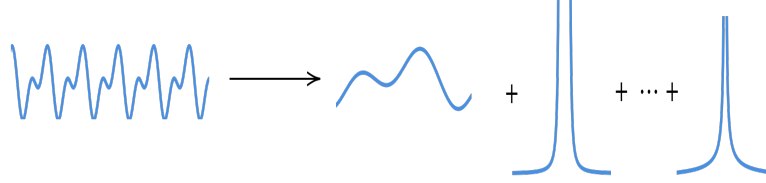


FIGURE 1. An illustration of the convergence occurring in Theorem 1.

For historical accuracy, we note that the argument used by Mercuri-Willem in [5] to prove Theorem 1 has its roots in a paper of Brézis-Coron [3] and was later refined in a book of M. Willem [11].

REMARK. For simplicity, we momentarily take $\mu = 1$. We then have a complete classification of the bubbles. Namely, they all take the form

$$\left[\frac{\lambda^{\frac{1}{p-1}} N^{\frac{p-1}{p^2}} \left(\frac{N-p}{p-1} \right)^{1/p'}}{\lambda^{\frac{p}{p-1}} + |x - x_0|^{\frac{p}{p-1}}} \right]^{\frac{N-p}{p}} \quad \text{for some } x_0 \in \mathbb{R}^N \text{ and } \lambda > 0.$$

This classification was established by

- Caffarelli-Gidas-Spruck in 1989 for $p = 2$;
- Damascelli-Merchán-Montoro-Sciunzi in 2014 for $\frac{2N}{N+2} \leq p < 2$;
- Vétois 2016 & Damascelli-Ramaswamy 2001 for $1 < p < \frac{2N}{N+2}$;
- Sciunzi 2015 for $2 \leq p < N$.

Those interested in a more detailed account of this classification should consult Sciunzi [7].

1.1. Some Properties of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. The homogeneous Sobolev space $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is really the “right” setting for dealing with critical equations of this type. The main reason lies within the GNS-inequality

$$\|u\|_{L^{p^*}(\mathbb{R}^N)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^N)}$$

which holds for all $u \in C_c^\infty(\mathbb{R}^N)$. By a simple calculation, it can be shown that both sides of this inequality are invariant under the transformation

$$u(x) \mapsto \lambda^{(N-p)/p} u(\lambda x), \quad \forall \lambda > 0.$$

By temporarily giving $\mathcal{D}^{1,p}(\mathbb{R}^N)$ the norm

$$\|u\|_* := \|u\| + \|u\|_{L^{p^*}(\mathbb{R}^N)},$$

it is easy to see that $C_c^\infty(\mathbb{R}^N)$ is dense in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Thus, by density, the GNS-inequality must hold for all $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. This then ensures that the norm $\|\cdot\|_*$ is equivalent to the standard norm $\|\cdot\|$. It then follows that $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is Banach. If (u_n) is Cauchy in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, we can find $v^0 \in L^{p^*}(\mathbb{R}^N)$ and $v^i \in L^p(\mathbb{R}^N)$ such that

$$\lim u_n = v^0 \quad \text{and} \quad \lim \partial_i u_n = v^i$$

in $L^{p^*}(\mathbb{R}^N)$ and $L^p(\mathbb{R}^N)$, respectively. Then, for $\varphi \in C_c^\infty(\mathbb{R}^N)$ we have

$$\int_{\mathbb{R}^N} v^0 \partial_i \varphi = \lim \int_{\mathbb{R}^N} u_n \partial_i \varphi = - \lim \int_{\mathbb{R}^N} \varphi \partial_i u_n = \int_{\mathbb{R}^N} \varphi v^i$$

whence we infer that $v^0 \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that $u_n \rightarrow v^0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$.

1.2. A Useful Convergence Criterion. The following result will be implicitly assumed throughout the proof of Theorem 1.

THEOREM 2. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let f_n be a sequence of measurable functions converging a.e. to a measurable function f on X . If $\{f_n\}_{n \geq 1}$ is uniformly bounded in $L^p(X, \mu)$ for some $p \in (1, \infty)$, then*

$$\int_X |(f_n - f)g| \, d\mu \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

for every $g \in L^{p'}(X, \mu)$.

Therefore, when $\mu(X) < \infty$, boundedness in $L^p(X, \mu)$ (with $p > 1$) ensures that $f_n \rightarrow f$ in $L^1(X, \mu)$, without the need for a dominating function.

2. Sketch of Proof of Theorem 1

STEP 1. It is a general fact that (PS) -sequences are bounded. Passing to a subsequence, $u_n \rightharpoonup v_0$ in weakly $W_0^{1,p}(\Omega)$ and strongly in $L^p(\Omega)$, u_n converges pointwise almost everywhere to v_0 on Ω , and $\phi'(v_0) = 0$. Furthermore, the sequence

$$u_n^1 := u_n - v_0 \in W_0^{1,p}(\Omega)$$

is bounded and satisfies

- (1) $\nabla u_n^1 \rightarrow 0$ almost everywhere on Ω ;
- (2) $\|u_n^1\|^p = \|u_n\|^p - \|v_0\|^p + o(1)$;
- (3) $\lim_{n \rightarrow \infty} \phi_\infty(u_n^1) = c - \phi(v_0)$; and
- (4) $\phi'_\infty(u_n^1) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

STEP 2. We are done if $u_n^1 \rightarrow 0$ strongly in $L^{p^*}(\Omega)$. Indeed, by Step 1, we know that $\phi'_\infty(u_n^1)$ converges to 0 in $W^{-1,p'}(\Omega)$. Thus,

$$\langle \phi'_\infty(u_n^1), u_n^1 \rangle = \int_{\mathbb{R}^N} (|\nabla u_n^1|^p - \mu |u_n^1|^{p^*}) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Because $u_n^1 \rightarrow 0$ in $L^{p^*}(\Omega)$, we infer that

$$\int_{\mathbb{R}^N} |\nabla u_n^1|^p \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

In other words, $\nabla u_n \rightarrow \nabla v_0$ in $L^p(\Omega)$. It follows that

$$\|u_n - v_0\| \rightarrow 0 \quad \text{and} \quad \|u_n\| \rightarrow \|v_0\|$$

as $n \rightarrow \infty$. Since $u_n \rightarrow v_0$ in $L^p(\Omega)$ by Step 1, we get $u_n^1 \rightarrow 0$ in $W_0^{1,p}(\Omega)$. That is,

$$\lim_{n \rightarrow \infty} u_n = v_0 \quad \text{strongly in } W_0^{1,p}(\Omega)$$

and it is then easy to show that $\phi(v_0) = \lim_{n \rightarrow \infty} \phi(u_n) = c$. By Step 1, v_0 solves (1.1) and is non-negative. Theorem 1 then follows with $k = 0$.

STEP 3. For every $n \geq 1$,

$$Q_n(r) := \sup_{y \in \bar{\Omega}} \int_{B(y,r)} |u_n^1|^{p^*}.$$

Each Q_n is continuous since $\left\{ \int_{B(y,\cdot)} |u_n^1|^{p^*} : y \in \bar{\Omega} \right\}$ is equibounded and locally equicontinuous on $[0, \infty)$. By Step 2, assume $u_n^1 \not\rightarrow 0$ in $L^{p^*}(\Omega)$. After passing to a subsequence, this means that there exists $\delta > 0$ such that

$$\inf_{n \in \mathbb{N}} \int_{\Omega} |u_n^1|^{p^*} > \delta.$$

Without loss of generality, we may assume that

$$0 < \delta < \left(\frac{S_p}{2\mu} \right)^{N/p} \tag{2.1}$$

where $S_p > 0$ is the best constant for which the Gagliardo-Nirenberg-Sobolev inequality holds in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Now, every Q_n is a continuous function on $[0, \infty)$ satisfying both

$$Q_n(0) = 0 \quad \text{and} \quad \lim_{r \rightarrow \infty} Q_n(r) > \delta.$$

Using the intermediate value and dominated convergence theorems,

$$\delta = \int_{B(y_n^1, \lambda_n^1)} |u_n^1|^{p^*}, \quad \forall n \in \mathbb{N} \text{ and some } (y_n^1, \lambda_n^1) \in \bar{\Omega} \times \mathbb{R}_+$$

STEP 4. For each $n \in \mathbb{N}$ consider the domain

$$\Omega_n := \frac{1}{\lambda_n^1} (\Omega - y_n^1).$$

A simple calculation shows that

$$x \in \Omega_n \iff \lambda_n^1 x + y_n^1 \in \Omega.$$

Next, we define a sequence $(v_n^1)_{n \in \mathbb{N}}$ in $W_0^{1,p}(\Omega_n) \subseteq W^{1,p}(\mathbb{R}^N)$ by the rule

$$v_n^1(x) := (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1).$$

The family $\{v_n^1\}_{n \in \mathbb{N}}$ belongs to the space $W^{1,p}(\mathbb{R}^N)$ and is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Moreover, one has

$$\delta = \sup_{y \in \bar{\Omega}_n} \int_{B(y,1)} |v_n^1|^{p^*} = \int_{B(0,1)} |v_n^1|^{p^*}. \quad (2.2)$$

For each $n \in \mathbb{N}$, $\phi'_\infty(u_n^1)$ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$ restricts to an element of $W^{-1,p'}(\Omega)$. We can then find $f_n^1, \dots, f_n^N \in L^{p'}(\Omega)$ such that

$$\langle \phi'_\infty(u_n^1), h \rangle = \sum_{i=1}^N \int_{\Omega} f_n^i \partial_i h, \quad \forall h \in W_0^{1,p}(\Omega).$$

Next, consider the functions given by

$$g_n^i = (\lambda_n^1)^{N-\frac{N}{p}} f_n^i(\lambda_n^1 x + y_n^1), \quad \forall i = 1, \dots, N.$$

STEP 5. Adhering to the notation and terminology developed above, one has

$$\langle \phi'_\infty(v_n^1), h \rangle = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h, \quad \forall h \in W_0^{1,p}(\Omega_n).$$

Now, passing to a subsequence if necessary, we may assume that $\lambda_n^1 \rightarrow \lambda^1 \in \mathbb{R}$ and that $y_n^1 \rightarrow y^1 \in \bar{\Omega}$. Thus, we may assume that $\Omega_n \rightarrow \Omega_\infty$ as $n \rightarrow \infty$, where Ω_∞ is

- a rescaled translation of Ω if $\lambda^1 > 0$;
- all of $\mathbb{R}^N$ if $\lambda^1 = 0$ and $y^1 \in \Omega$; or
- a half space if $\lambda^1 = 0$ and $y^1 \in \partial\Omega$.

STEP 6. After passing to a subsequence, we may assume that $v_n^1 \rightharpoonup v_1$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$;

$$\lim_{n \rightarrow \infty} v_n^1(x) = v_1(x) \quad \text{a.e.}$$

We use the reflexivity of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and the boundedness of $\{v_n^1\}$ by homogeneity. As for a.e. convergence, we use a diagonal argument with successive applications of the Rellich-Kondrachov theorem on a sequence of open balls $\{B_k\}_{k \geq 1}$ exhausting $\mathbb{R}^N$. Then, Fatou's

lemma gives the assertion.

STEP 7. The weak $\mathcal{D}^{1,p}(\mathbb{R}^N)$ -limit v_1 of v_n^1 is non-zero.

We argue by contradiction. By taking a sequence of open balls $B_k \nearrow \mathbb{R}^N$ and using the Rellich-Kondrachov theorem, we may assume that $v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$.

By using Step 1 together with a duality result from Brézis [2] (Proposition 9.20 therein), it follows that

$$\|\phi'_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} = \sum_{i=1}^N \int_{\Omega} |f_n^i|^{p'} \quad (2.3)$$

$$= \sum_{i=1}^N \int_{\Omega_n} |g_n^i|^{p'} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (2.4)$$

If $y \in \Omega_\infty$, then y belongs to Ω_n for all sufficiently large n . Fix $y \in \Omega_\infty$ and consider the ball $B(y, 1)$. For a given test function $h \in C_c^\infty(\mathbb{R}^N)$ with $\text{supp}(h) \subset B(y, 1)$, Hölder's inequality and Sobolev's inequality gives

$$\int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} \leq S_p^{-1} \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |\nabla(hv_n^1)|^p \right). \quad (2.5)$$

Because v_n^1 converges *strongly* to 0 in $L_{\text{loc}}^p(\mathbb{R}^N)$, one can show that

$$\int_{\Omega_n} |\nabla(hv_n^1)|^p = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i(|h|^p v_n^1) + \mu \int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} + o(1) \quad (2.6)$$

$$\leq \mu \int_{B(y,1)} |h|^p |v_n^1|^{p^*} + o(1) \quad (2.7)$$

$$\leq \mu S_p^{-1} \delta^{\frac{p}{N}} \int_{\mathbb{R}^N} |\nabla(hv_n^1)|^p + o(1) \quad (2.8)$$

$$\leq \frac{1}{2} \int_{\mathbb{R}^N} |\nabla(hv_n^1)|^p + o(1) \quad (2.9)$$

Whence we conclude that

$$\lim_{n \rightarrow \infty} \int_{\Omega_n} |\nabla(hv_n^1)|^p = 0.$$

This means that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(B)$, for any ball B of radius 1 with center in Ω_∞ . To summarize, we have shown that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$ whence $v_n^1 \rightarrow 0$ in $W_{\text{loc}}^{1,p}(\mathbb{R}^N)$. Finally, the Sobolev inequality implies that

$$\lim_{n \rightarrow \infty} v_n^1 = 0 \quad \text{in } L_{\text{loc}}^{p^*}(\mathbb{R}^N),$$

which directly contradicts (2.2).

STEP 8. The radii converge to 0 as $n \rightarrow \infty$, i.e. $\lambda^1 = \lim_{n \rightarrow \infty} \lambda_n^1 = 0$. Arguing by contradiction, let us assume that $\lambda^1 > 0$. Let $\varphi \in C_c^\infty(\mathbb{R}^N; \mathbb{R}^N)$ and note that

$$\int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi = (\lambda_n^1)^{N/p-N} \int_{\mathbb{R}^N} \nabla u_n^1(z) \cdot \varphi \left(\frac{z - y_n^1}{\lambda_n^1} \right) dz.$$

Now, Step 1 ensures that $\nabla u_n^1 \rightarrow 0$ pointwise almost everywhere on $\mathbb{R}^N$. Since $\lambda_n^1 \rightarrow \lambda^1 > 0$ and $y_n^1 \rightarrow y^1 \in \bar{\Omega}$, there exists a compact set $\Lambda \subset \mathbb{R}^N$ such that

$$\text{supp} \left(\varphi \left(\frac{\cdot - y_n^1}{\lambda_n^1} \right) \right) \subseteq \Lambda, \quad \forall n \geq 1.$$

Using now that φ is bounded,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla v_n^1 \cdot \varphi = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \nabla u_n^1(z) \cdot \varphi \left(\frac{z - y_n^1}{\lambda_n^1} \right) dz = 0.$$

By a density argument, we find that $v_n^1 \rightarrow 0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, which is a contradiction.

STEP 9. v^1 is non-negative almost everywhere on $\mathbb{R}^N$. Since $v_n^1 \rightarrow v^1$ a.e., this amounts to showing that $\lim_{n \rightarrow \infty} v_n^1 \geq 0$ almost everywhere. To see this, write

$$v_n^1(x) = (\lambda_n^1)^{(N-p)/p} u_n(\lambda_n^1 x + y_n^1) - \underbrace{(\lambda_n^1)^{(N-p)/p} v_0(\lambda_n^1 x + y_n^1)}_{=: w_n}$$

where, by homogeneity, the negative part of $(\lambda_n^1)^{(N-p)/p} u_n(\lambda_n^1 x + y_n^1)$ tends to 0 in $L^{p^*}(\mathbb{R}^N)$ as $n \rightarrow \infty$. Now, for all n sufficiently large, we have

$$\int_{B_R(0)} |w_n|^{p^*} = \int_{B_{R\lambda_n^1}(y_n^1)} |v_0|^{p^*} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This means that $w_n \rightarrow 0$ strongly in $L^{p^*}(B_R)$ for each $R > 0$ fixed. Passing to a subsequence completes the proof.

In light of this, exactly one of the two following occurs:

- (1) $y^1 \in \Omega$ and $\Omega_\infty = \mathbb{R}^N$; or
- (2) $y^1 \in \partial\Omega$ and Ω_∞ is a half-space.

In the second case, by an argument similar to that given in Lemma 2, the limit v^1 can be seen to be a $\mathcal{D}_0^{1,p}(H)$ -solution to the problem

$$\begin{cases} -\Delta_p u \equiv \mu u^{p^*-1} & \text{in } H \\ u \geq 0 & \text{in } H. \end{cases}$$

Since $v^1 \neq 0$, this contradicts Theorem 1.1 in Mercuri-Willem [5]. Hence, the only possibility is that $\Omega_\infty = \mathbb{R}^N$ and so we may assume that (y_n^1) belongs to Ω for each $n \in \mathbb{N}$.

STEP 10. Let $w \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ be a non-trivial critical point of ϕ_∞ . Then,

$$0 = \langle \phi'_\infty(w), w \rangle = \int_{\mathbb{R}^N} (|\nabla w|^p - \mu |w|^{p^*})$$

whence

$$S_p \|w\|_{L^{p^*}}^p \leq \|\nabla w\|_{L^p(\mathbb{R}^N; \mathbb{R}^N)}^p = \|w\| = \mu \|w\|_{L^{p^*}(\mathbb{R}^N)}^{p^*}.$$

From this, we obtain

$$\phi_\infty(w) \geq \frac{\mu}{N} \left(\frac{S_p}{\mu} \right)^{N/p} > 0.$$

Through a technical argument similar to that in Step 1, the sequences

$$u_n^2(x) := u_n^1(x) - (\lambda_n^1)^{\frac{p-N}{N}} v_1 \left(\frac{x - y_n^1}{\lambda_n^1} \right)$$

satisfies

- (1) $\|u_n^2\|^p = \|u_n\|^p - \|v_0\|^p - \|v_1\|^p + o(1)$;
- (2) $\phi_\infty(u_n^2) \rightarrow \lim_{n \rightarrow \infty} \phi(u_n) - \phi(v_0) - \phi_\infty(v_1)$; and
- (3) $\phi'_\infty(u_n^2) \rightarrow 0$ in $W^{-1,p'}(\Omega)$.

Then using Step 10, the previously outlined procedure can only iterate finitely many times and the proof will be complete once we find ourselves in Step 2. This completes the proof.

Appendix A. Technical Tools

LEMMA 1. Let (u_n) be a sequence in $W_0^{1,p}(\Omega)$ such that as $n \rightarrow \infty$:

- (1) $u_n \rightharpoonup u$ in $W_0^{1,p}(\Omega)$;
- (2) $u_n \rightarrow u$ pointwise a.e. on Ω ;
- (3) $\|(u_n) - \|_{L^{p^*}(\Omega)} \rightarrow 0$;
- (4) $\phi(u_n) \rightarrow c$ and $\phi'(u_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$.

After passing to a subsequence, we have $\nabla u_n \rightarrow \nabla u$ pointwise a.e. on Ω and $\phi'(u) = 0$, i.e. $u \geq 0$ solves (1.1). Furthermore, the sequence $v_n := u_n - u$ in $W_0^{1,p}(\Omega)$ satisfies

- (i) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|v_n\|^p) = \|u\|^p$;
- (ii) $\phi_\infty(v_n) \rightarrow c - \phi(u)$; and
- (iii) $\phi'_\infty(v_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$.

REMARK. The heart is the a.e. convergence of the gradients. Afterwards, everything will follow from a combination of the Brézis-Lieb lemma and L^p -convergence results.

We now tackle an analogue of Lemma 1 for the limiting problem (1.3).

LEMMA 2. Let (y_n) and (λ_n) be sequences in $\bar{\Omega}$ and $\mathbb{R}_+$, respectively and assume that

$$\lim_{n \rightarrow \infty} \frac{\text{dist}(y_n, \partial\Omega)}{\lambda_n} = \infty. \quad (\text{A.1})$$

Let (u_n) be a sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and define

$$v_n(x) := \lambda_n^{\frac{(N-p)}{p}} u_n(\lambda_n x + y_n).$$

Suppose that, as $n \rightarrow \infty$,

- (1) $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$;
- (2) $v_n \rightarrow v$ pointwise a.e. on $\mathbb{R}^N$;
- (3) $(u_n)_- \rightarrow 0$ in $L^{p^*}(\mathbb{R}^N)$;
- (4) $\phi_\infty(u_n) \rightarrow c$ and $\phi'_\infty(u_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$.

Then, $v \geq 0$ a.e. on $\mathbb{R}^N$, $\phi'_\infty(v) = 0$, and, after passing to a subsequence, $\nabla v_n \rightarrow \nabla v$ pointwise a.e. on $\mathbb{R}^N$. Moreover,

- (i) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|w_n\|^p) = \|v\|^p$;
- (ii) $\phi_\infty(w_n) \rightarrow c - \phi_\infty(v)$ as $n \rightarrow \infty$; and
- (iii) $D\phi_\infty(w_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$

where we define

$$w_n(z) := u_n(z) - \lambda_n^{\frac{p-N}{N}} v\left(\frac{z - y_n}{\lambda_n}\right).$$

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